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SECTOR AT A GLANCE 03

Disease Programmes

Health Sector in India

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Disease Programmes

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The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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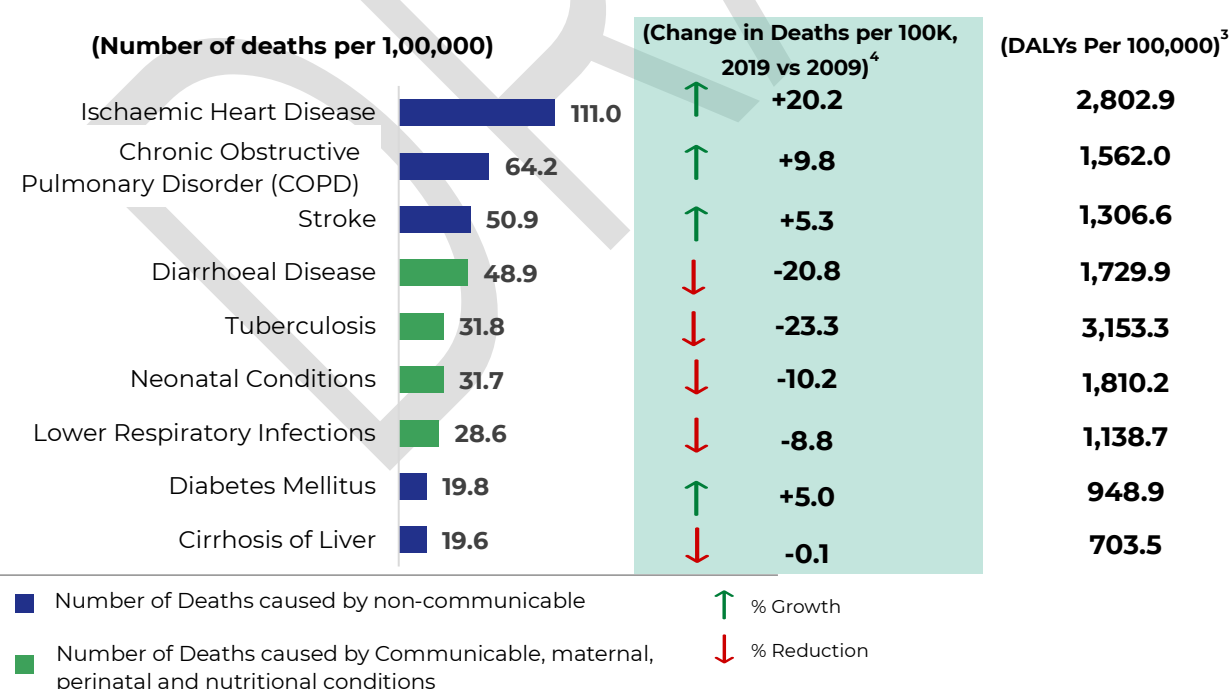
Overview

India has embarked on a journey to provide universal health coverage (UHC) to its citizens with the National Health Policy, 2017 and the introduction of the Pradhan Mantri Jan Arogya Yojana (PMJAY) scheme for the poorest and vulnerable populations in 2018.¹ As India makes rapid strides in its social and economic development, a significant epidemiological transition is also underway. India faces increasing burdens of non-communicable diseases and changes in its demographic profile which has consequences on health investments and delivery. India is undergoing four crucial transitions in demography, disease patterns, donor spending on health, and domestic health financing (the “4Ds”) which will have a profound impact on the country’s ability to move towards UHC. In the face of these transitions, there are many challenges that the country needs to tackle to successfully achieve UHC.

In the last 26 years, there has been a noticeable change in the disease trends in the country, marked by a significant reduction in mortality attributed to communicable, maternal, neonatal, and nutritional diseases (CMNNDs). This decline has led to an increase in the average lifespan of India's population. There is an increase in the share of ageing populations with the proportion of people over the age of 65 years expected to double to 14% by 2050. One third of India's population is in the 10-24 years age group and the working age population is expected to grow by roughly 9.7 million per year during 2021-2031. Non-communicable diseases have become more prominent like global trends. India's youth face an increasing burden of non-communicable diseases (NCDs), including mental health disorders, emerging as leading causes of death and accounting for 60% of total deaths in 2016 compared to 38% in 1990. The three NCDs that emerged as the leading causes of death in 2017 are ischemic heart disease, chronic obstructive pulmonary disease (COPD), and stroke.

This scenario presents a dual challenge for India's healthcare system. While the absolute burden of diseases like diarrhoea, lower respiratory infections, tuberculosis, and neonatal disorders is decreasing, it still remains considerable. Moreover, the impact of non-communicable conditions such as heart disease, stroke, and diabetes on health is growing. The nature of this challenge, however, varies across different regions of the country.²

Fig 1 - Top Causes of Death and their DALYs in India, 2019^{3,4}



DALYs: Disability-adjusted life year are the sum of the years of life lost to due to premature mortality and the years lived with a disability due to prevalent cases of the disease

¹ Centre for Policy Impact. (2022). *India's Health Transitions*.

² Health Data. (2017). *India: Health of the Nation's States*.

³ World Health Organization. (n.d.). *Global Health Estimates - Leading Causes of Death*. Retrieved 2 September 2023.

⁴ Christopher J L Murray. (2020). Global burden of 369 diseases and injuries in 204 countries and territories, 1990–2019: a systematic analysis for the Global Burden of Disease Study.

- Tracking annual mortality rates is essential for identifying the underlying reasons for deaths and adjusting healthcare systems to respond efficiently. It also aids in understanding people's lifestyles to enhance healthcare services and minimize avoidable fatalities in all nations, effectively responding to evolving epidemiological conditions.
- Proportion of deaths in India due to several diseases has changed in 2019 as compared to 2009. Non-communicable diseases have become more prominent with Ischaemic Heart Disease (IHD) and COPD being the top causes of death and IHD as the leading contributor to mortality.
- Meanwhile, in 2019, mortality due to communicable diseases such as diarrhoeal disease, tuberculosis and lower respiratory tract disorders has declined in comparison to 2009.

Since India became independent, several measures have been undertaken by the national government to improve the health of the people. Prominent among these measures are the national health programmes which have been launched by the central government for the control and eradication of communicable diseases, improvement of environmental sanitation, raising the standard of nutrition, control of population and improving rural health. Various international agencies like World Health Organization (WHO), United Nation's Children Fund (UNICEF), United Nations Population Fund (UNFPA), World Bank and several foreign agencies like Swedish International Development Cooperation Agency (SIDA), Danish International Development Agency (DANIDA) and United States Agency for International Development (USAID) have been providing technical and material assistance in the implementation of these programmes. A brief account of the evolution of these programmes which are currently in operation is given in the following sections. [Park, K. (2021) *Park's Textbook Of Preventive And Social Medicine*].

Evolution of Disease Programmes in India

The public health structure in India has evolved over time, with various programmes and initiatives being implemented to address the country's health needs. The **Bhore Committee Report of 1946** was a landmark report that led to the development of the current health policy and systems in India.⁵

Post-independence, it was not until 1983 that the first health policy was formulated and adopted. Before that, healthcare initiatives were implemented under the Five-Year Plans, each with specific objectives. Finally, in **1983, National Health Policy** was introduced, aiming to achieve universal healthcare that is affordable and tailored to the needs of the population.⁵

1983 NHP Salient Features:-

- It emphasized the need for establishing comprehensive primary health care services to reach the population in remote areas of the country.
- It stressed on preventive, public health, and rehabilitation aspects of healthcare.
- It aimed to make health services and medicine affordable and sought to prevent disparity in social, economic, or current health status.
- It aimed to promote community participation in the planning and implementation of health programmes.

5. Grover, A. et al. (2019). *Health Policy, Programmes and Initiatives*.

In **2002, the NHP document** was released with the objective of achieving acceptable standards of health. It emphasized decentralization, equity, accessibility of healthcare services, and the provision of affordable private healthcare⁵

2002 NHP Salient Features:-

- It aimed to strengthen human resources in the health sector, including training and capacity building of health care providers. It stressed on preventive, public health, and rehabilitation aspects of healthcare.
- Main components of the 2002 NHP applied equally to the alternative systems of medicines, with specific features for these systems.
- It aimed to consolidate all health programmes under a single, overarching framework, expanding and improving public health programmes.
- It emphasized an increased allocation of 55 percent of the total public health investment for the primary health sector, with secondary and tertiary health sectors also receiving attention.

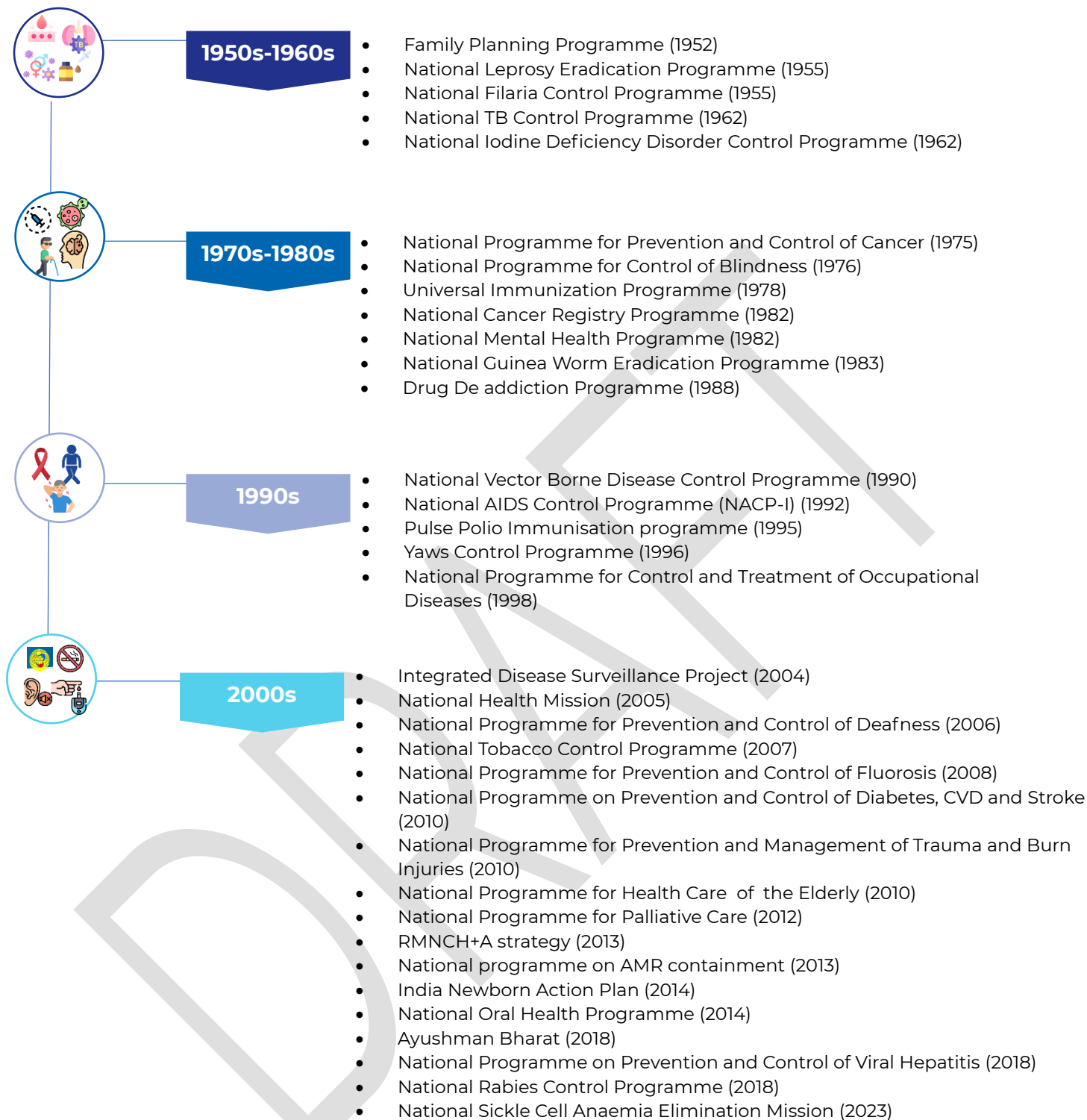
In **2017, the National Health Policy** was **updated** after a 14-year gap, reflecting the changing health landscape. It acknowledged the rising prevalence of non-communicable and infectious diseases, the growth of the private healthcare sector, increased healthcare expenditure, etc.⁵

2017 NHP Salient Features:-

- Landscape of health priorities is evolving. While there has been a significant reduction in maternal and child mortality rates, there is a mounting burden associated with noncommunicable diseases and certain infectious diseases.
- Development of a strong healthcare sector, projected to experience double-digit growth
- Increasing occurrences of financial hardships caused by healthcare expenses, currently identified as a significant contributor to poverty.
- Increasing economic growth provides the opportunity for improved fiscal capacity. Consequently, there is a need for a new healthcare policy that can adapt to these changing circumstances.

Apart from the National Health Policies (NHPs), various other policy measures have been periodically introduced, all closely interconnected with the aim of enhancing the health and well-being of the population. Some notable examples include the National Population Policy, National Nutrition Policy, National Water Policy, and National Environmental Policy, among others.⁵

Timeline of Introduction of Disease Programmes in India



National Health Programmes



The Ministry of Health and Family Welfare is implementing various schemes and programmes and national initiatives to provide universal access to quality health care. The approach is to increase access to the decentralised health system by establishing new infrastructure in deficient areas and by upgrading the infrastructure in existing institutions. As part of the plan process, many different programmes have been brought together under the overarching umbrella of **National Health Mission** with NRHM and NUHM as its two sub-Missions.⁶



The **RMNCH+A Strategy** is framework based on provision of comprehensive care through the five pillars, or thematic areas, of reproductive, maternal, neonatal, child and adolescent health, and is guided by central tenets of equity, universal care, entitlements, and accountability. The aim is to reach the maximum number of people in the remotest corners of the country through a continuum of services, constant innovation and routine monitoring of interventions.⁷



India Newborn Action Plan: A targeted strategy for accelerating the reduction of preventable newborn deaths and still births in the country with the goal to attain “Single Digit Neonatal Mortality Rate” by 2030” and “Single Digit Still births Rate by 2030.”⁸



Family Planning Programme: India was the first country in the world to have launched a National Programme for Family Planning in 1952. Over the decades, the programme has undergone transformation in terms of policy and actual programme implementation and currently being repositioned to not only achieve population stabilization goals but also promote reproductive health and reduce maternal, infant and child mortality and morbidity.⁹



Universal Immunization Programme India's Immunization programme is one of the largest public health programmes in the world. Launched as Expanded programme on Immunization in 1978, it was renamed as Universal Immunization programme in 1985 when it was expanded beyond urban areas. It targets 3.04 crore pregnant women and 2.7 crore new borns annually. More than 1.2 crore immunization sessions are being conducted annually. Under UIP, immunization is being provided free of cost against 12 vaccine preventable diseases: Nationally against 11 diseases: Diphtheria, Pertussis, Tetanus, Polio, Measles, Rubella, severe form of Childhood Tuberculosis, Rotavirus diarrhoea, Hepatitis B, Meningitis and Pneumonia caused by Hemophilus Influenza type B and Pneumococcal Pneumonia and sub-nationally against one disease - Japanese Encephalitis (JE vaccine is provided only in endemic districts). It is the most cost-effective public health intervention, largely responsible for reduction of vaccine preventable diseases and, thus, contributing to decrease in India's Under 5 mortality rates from 45 per 1,000 live births in 2014 to 35 per 1,000 live births in 2019.¹⁰



Ayushman Bharat is an attempt to move from a selective approach to health care to deliver comprehensive range of services spanning preventive, promotive, curative, rehabilitative and palliative care. It has two components which are complementary to each other. Under its first component, 1,50,000 Health and Wellness Centres (HWCs) will be created to deliver Comprehensive Primary Health Care, that is universal and free to users, with a focus on wellness and the delivery of an expanded range of services closer to the community. The second component is the Pradhan Mantri Jan Arogya Yojana (PM-JAY) which provides health insurance cover of Rs. 5 lakh per year to over ten crore poor and vulnerable families for seeking secondary and tertiary care.¹¹



National Programme for Prevention and Control of Non-Communicable Diseases [NP-NCD] (previously known as National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke[NPCDCS]): India is experiencing a rapid health transition with a rising burden of Non-Communicable Diseases (NCD) surpassing the burden of Communicable diseases like water-borne or vector borne diseases, TB, HIV, etc. The Non-Communicable Diseases like Cardiovascular diseases, Cancer, Chronic Respiratory Diseases, Diabetes, etc. are estimated to account for around 60% of all deaths. The primary objective of programme is to address the growing burden of non-communicable diseases through prevention, early detection, and management. The program aims to promote healthier lifestyles, reduce risk factors, and improve access to NCD healthcare services nationwide.¹²

6 National Health Mission. (n.d.). *About Us*. Retrieved 2 September 2023.

7 NHM. (n.d.). *RMNCH+A*. Retrieved 3 September 2023.

8 NHM. (n.d.). *INAP*. Retrieved 3 September 2023.

9 NHM. (n.d.). *FPP*. Retrieved 3 September 2023.

10 Sample Registration System (SRS). (2020). *Statistical Report*.

11 National Health Authority. (n.d.). *Ayushman Bharat*. Retrieved 3 September 2023.

12 MoHFW (2023). *Annual Report 2022-23*.



Integrated Disease Surveillance Project is a decentralised state-based surveillance system in the country. The project is intended to detect early warning signals of impending outbreaks and help initiate an effective response in a timely manner in urban and rural areas. The key objective of the programme is to strengthen/maintain decentralized laboratory-based IT enabled disease surveillance system for epidemic prone diseases to monitor disease trends and to detect and respond to outbreaks in early rising phase through trained Rapid Response Team (RRTs).¹³



National Programme for Mental Health: The goal is to secure the presence and reach of essential mental healthcare for everyone in the foreseeable future, with a special emphasis on extending it to the most marginalized segments of society. Additionally, the aim is to promote the integration of mental health insights into mainstream healthcare practices and broader social advancement efforts.¹²



National Programme for Control of Blindness and Visual Impairment: focuses on preventing avoidable blindness and providing quality eye care services across India. Through screening, treatment, and awareness, the programme aims to reduce the burden of visual impairment in the population.¹²



National Programme for Prevention and Control of Deafness: The objective is to avert unnecessary hearing impairment caused by disease or injury, by promptly detecting, diagnosing, and addressing ear-related issues that lead to hearing loss and deafness.¹²



National Oral Health Programme: It is dedicated to improving oral health across India through preventive measures, treatment access, and oral health awareness initiatives. It aims to enhance overall oral well-being of the population by integrating dental care into the broader healthcare frameworks.¹²



National Programme for Prevention and Management of Trauma and Burn Injuries: This initiative was implemented to lower the occurrence of burn injuries, decrease mortality rates, alleviate morbidity, and reduce disability stemming from such incidents. Its primary objectives include enhancing awareness.¹²



National Programme for the Health Care of Elderly was launched with objective of tackling diverse health challenges faced by the elders. This was conceived to ensure the elderly population's access to reasonably priced, easily reachable and top-notch long-term care services, while also focusing on a comprehensive approach.¹²



National AIDS Control Programme was launched in India in the year 1987. The Ministry of Health and Family Welfare has set up National AIDS Control Programme Organization as a separate wing to implement and closely monitor the various components of the programme. The aim of the programme is to prevent further transmission of HIV, to decrease morbidity and mortality associated with HIV infection and to minimize the socio-economic impact resulting from HIV infection.¹²

¹³ MoHFW. (n.d.). *IDSP*. Retrieved 3 September 2023.

Disease Control Programmes (under NHM)



National Vector Borne Diseases Control Programme

It serves as a comprehensive programme aimed at preventing and managing vector-borne illnesses such as Malaria, Japanese Encephalitis, Dengue, Chikungunya, Kala-azar, and Lymphatic Filariasis.¹²



National Leprosy Eradication Programme

Programme offers both technical and financial support to states for all aspects of dealing with leprosy, including detection, diagnosis, treatment, and post-treatment care for affected individuals.¹²



National TB Elimination Programme

It Initiated with a mission-oriented goal of attaining TB eradication by 2025, this programme employs a multifaceted strategy to identify all TB cases.¹²



National Iodine Deficiency Disorder Control Programme

It focuses on ensuring the adequate intake of iodine through salt fortification, promoting awareness about the importance of iodized salt, and preventing iodine deficiency.¹²

Disease Trends in India

India is undergoing a swift health transformation, where the burden of Non-Communicable Diseases (NCDs) has exceeded that of Communicable diseases like water-borne or vector borne diseases, TB, HIV, etc. NCDs such as Cardiovascular diseases, Cancer, Chronic Respiratory Diseases, and Diabetes account for an estimated 60% of all deaths, leading to substantial loss in productive years.¹⁴ The nation has made remarkable strides in curtailing major communicable diseases, achieving worldwide smallpox eradication and eliminating polio within its borders. Additionally, India has significantly decreased HIV incidence by over 50% in the past two decades. Recent outbreaks that of Nipah virus and COVID-19 and have been effectively managed and contained.¹⁵

A. Communicable Disease

1. Tuberculosis

- The year 2022 also saw the notification of the highest number of TB cases at 24.2 lakhs, leading to an increase of the annualised TB case notification rate (ACNR) from 153 per lakh in 2021 to 172 per lakh population.¹⁶
- Incidence of Tuberculosis (in country model) – 196 per 1,00,000 population (all forms)¹⁸ and prevalence rate (National TB Prevalence Survey (NATBPS) 2019-2021) of 312 per 1,00,000 population. (all forms)¹⁷
- The country experienced an 11% uptick in the mortality rate from all types of TB between 2019 and 2020, indicating a modest rise.¹⁶

¹⁴ NHM. (n.d.). NHM Component. Retrieved 11 March 2024

¹⁵ India Science, technology and Innovation

¹⁶ MoHFW. (2022). India TB Report 2022.

¹⁷ MoHFW. (2021). TB Prevalence Survey 2021.

¹⁸ TB Facts India. (n.d.). TB Statistics India.

National TB Programme Achievements (2021):

- Among all TB patients who have been notified, 95% are aware of their HIV status.¹⁶
- More than 9,20,000 individuals underwent TB screening, with over 65,000 showing symptoms of TB, and over 2,300 received TB diagnoses.¹⁶
- Among those who displayed TB symptoms, more than 82% underwent at least one TB test.¹⁶
- 2,51,675 People living with HIV/AIDS given access to rapid molecular testing via CBNAAT for TB diagnosis.¹⁶
- The utilization of Bedaquiline (Bdq) has been extended to encompass children aged five and older who weigh 15 kg or more, as well as pregnant women. Additionally, Bdq can now be employed in specific patients beyond six months of age, and its concurrent administration with Delamanid has been introduced.
- 13,26,306 beneficiaries paid through Nikshay Poshan Yojana.¹⁶

2. HIV



- 0.21% - Prevalence of adult (15–49 years) HIV/AIDS, 2021.¹⁹
- 0.05 - Incidence of per 1000 uninfected population HIV/AIDS, 2021.¹⁹
- From 2010 to 2021, there was a 46.3% decline in India's annual rate of new HIV infections.¹⁹
- Over the last two decades, adult HIV prevalence for 15 to 49 years has decreased significantly to 0.2% in 2021; since reaching its peak of 0.5% in 2000.¹⁹

Incidence rate is showing downward trend with the exception of some northeast states (Mizoram, Nagaland, Meghalaya, and Manipur).¹⁹

Nations AIDS Control Programme:

- In 2022, "National Guidelines for HIV Care and Treatment 2021" mobile APP was launched.¹²
- National average for 12-month retention was 71% in 2020-2021, and it increased to 80% in 2021-2022.¹²
- ART initiation rate saw an improvement, rising to 95% in January-September 2022, up from 92% during the same period in 2021.¹²
- As of April 2020, India began prescribing Dolutegravir-based ARV regimens for PLHIV in line with WHO and ART TRG recommendations. Currently, over 13.6 lakh patients, approximately 87% of all PLHIV on ART, have transitioned to the Dolutegravir (DTG) regimen.¹²
- Among patients co-infected with HIV and TB, 95.3% were started on Co-trimoxazole Preventive Therapy (CPT), 99% on Anti-Tubercular therapy, and 94.4% on Anti-Retroviral Therapy, as reported in MPR-Jan-Sep 2022.¹²

CBNAAT: Cartridge-based Nucleic Acid Amplification Test, TRG: Technical Resource Group on HIV Surveillance and Estimations

¹⁹ NACO. (2022). *India HIV Estimates 2021*.

²⁰ Anjana et al. (2023). *Metabolic Non-Communicable Disease Health Report of India*.

²¹ Ranasinghe P et al. (2021). *Prevalence and trends of the diabetes epidemic in urban and rural India*.

²² World Health Organization. (n.d.). *India Hypertension Control Initiative*.

²³ Dandona et al. (2018). *The changing patterns of cardiovascular diseases and their risk factors in the states of India*.

²⁴ International Institute for Population Sciences, National Programme for Health Care of Elderly, Ministry of Health and Family Welfare, Harvard T. H. Chan School of Public Health, and the University of Southern California. Et al. (2020). *The Longitudinal Ageing Study in India, Wave 1, 2017–18*.

B. Non-Communicable Disease

1. Diabetes



- 10.1 crore - Prevalence of Diabetes, 2023.²⁰
- 13.6 crore - Prevalence of Prediabetes, 2023.²⁰
- Occurrence of diabetes surged in both rural and urban regions of India, rising from 2.4% and 3.3% in 1972 to 15.0% and 19.0%, respectively, during the period spanning 2015 to 2019.²¹

Highest prevalence rates observed in Goa (26.4 %), Puducherry (26.3%), and Kerala (25.5%).²⁰

2. Cardiovascular Disease



- An estimated 20 crore adults in India suffer from hypertension, but less than 10% of them manage to keep their blood pressure in check.²²
- 53% percent increase in prevalence of Ischaemic Heart Disease between 1990 and 2016.²³

Prevalence of CVDs among elderly aged 60 was found to be higher in Goa (60%), Kerala (57%), and Chandigarh (55%).²⁴

3. Cancer



- A total of 8.5 lakh cancer-related deaths were reported in 2020 for all age groups.²⁵
- The incidence of cancer cases is estimated to increase by 12.8 % in 2025 as compared to 2020.²⁵

Uttar Pradesh, Maharashtra and West Bengal have the highest Estimated Incidence of Cancer Cases.²⁶

National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD)

- In the fiscal year 2022-23, following healthcare facilities were established: 696 District NCD Cells, 707 District NCD Clinics, 193 District Cardiac Care Units, 268 District Day Care Centers for cancer patients, and 5541 CHC NCD Clinics.¹²
- India Hypertension Control Initiative, a joint effort involving MoHFW, ICMR, State Governments, and WHO India, has been implemented in 141 districts across 25 states.¹²

²⁵ Sathishkumar, et al. (2022). Cancer incidence estimates for 2022 & projection for 2025.

²⁶ Data.gov. (n.d.). *State/UTs-wise Estimated Incidence of Cancer Cases in India*. Retrieved 5th September 2023.

²⁷ Lahariya C. (n.d.). 2020. Health & Wellness Centers to Strengthen Primary Health Care in India.

²⁸ Missionlife. (n.d.). *About Mission LIFE*. Retrieved 6 September 2023.

²⁹ Invest India. (2023). *Yoga Day 2023*. Retrieved 6 September 2023.

Opportunities and Challenges

A. Opportunities

AB-HWCs

- Ayushman Bharat (AB) is an attempt to move from a selective approach to health care to deliver comprehensive range of services spanning preventive, promotive, curative, rehabilitative and palliative care.
- HWCs are envisioned to provide an extended array of services that surpass maternal and child healthcare, encompassing non-communicable disease management, oral, eye, and ear care, mental health services, as well as primary care.²⁷

Mission LiFE

- "Mission LiFE" initiative encourages a healthy lifestyle that may be scaled to improve awareness about the lifestyle risk factors for NCDs.²⁸
- 1.6 crore Yoga/Wellness sessions were done at AB-HWC (as of 2022), which can be increased further by collaborating with Mission LiFE.²⁹

One Health

- Through One Health approach, India can effectively fulfill the requirement for streamlined disease reporting and enhance its capabilities to manage both economically significant illnesses and zoonotic diseases. Different ministerial departments have collaborated to achieve goals and vision of One Health.³⁰

Adjuvant Development Programme

- This programme is a collaboration between department of biotechnology and national Institute of allergy and infectious diseases (USA).
- It works to increase the effectiveness of the vaccines. Ex – Bharat Biotech's COVAXIN was developed with help under this programme.³¹

Leveraging Technology to Transform Public Health

- Modern technology holds significant promise in improving service efficiency and expanding the accessibility of healthcare and also enhancing the quality of services.
- Eg: Ni-kshay is a tuberculosis control programme designed to oversee and monitor activities and the current status of services pertaining to the screening, diagnosis, treatment, and follow-up of tuberculosis cases.³²

AYUSH

- From a policy standpoint, it's important to recognize that the utilization of AYUSH services is notably higher when it comes to addressing skin-related or musculoskeletal conditions.
- This trend indicates that individuals often turn to AYUSH when their treatment needs are long term. This pattern signifies the potential for enhancing the role of AYUSH care, especially in the management of chronic diseases.³³

B. Challenges

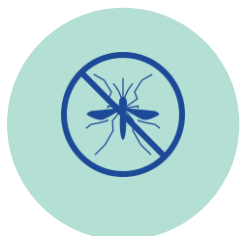


Rising life-style related disorders

- Risk factors for NCD are related to factors including lifestyle, food, exercise, stress, loss of urban green space, marketing of Ultra Processed Foods (UPFs) etc.
- More preventive schemes like Fit India movement, need to be promoted for raising awareness on lifestyle, food choices, etc.³⁴

Growing concern of AMR

- Antimicrobial resistance (AMR) is a growing public health problem.³⁵
- Even to newer and stronger antimicrobial medicines like carbapenems (broad spectrum antibiotics), resistance has developed.

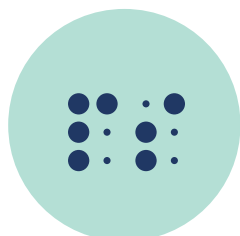


Resistant strains for vector-borne diseases

- A number of new infections like monkey pox³⁶ have emerged and pathogens and diseases like Dengue³⁸, Chikungunya³⁷, etc. have re-emerged with resistant or mutant strains.

Limited data on rare diseases

- Data (epidemiological) on how many people suffer from different diseases that are considered rare globally, is lacking in India and the cases identified so far have been diagnosed at tertiary hospitals.



Surveillance operates within distinct programme and institutional verticals

- Vertical programmes like the National AIDS Control Programme and the National TB Elimination Programme have made substantial strides in curbing disease transmission.
- Nevertheless, there is still a lack of complete integration of surveillance data from these vertical programmes into a unified surveillance platform. Furthermore, the utilization of existing data systems for addressing critical programmatic and policy inquiries, often referred to as "Research," has remained constrained.³⁵

30 Department OF Animal Husbandry & Dairying. (2022). *One Health Pilot Project*.

31 NIH. (2022). *Global Research in India*. Retrieved 7 September 2023.

32 National Government Services Portal. (n.d.). *Nikshay- Health establishment registration for TB notification*. Retrieved 7 September 2023.

33 Rudra S. (2017). Utilization of alternative systems of medicine as health care services in India, 2014.

34 Ministry of Youth Affairs and Sports. (n.d.). *Fit India Moment*. Retrieved 8 September 2023.

35 NITI. (2023). *Vision 2035, Public Health Surveillance In India*.

36 National Institute of Health & Family Welfare. (n.d.). *Daily News Bulletin*.

37 National vector Borne Disease Control. (2016). *National Guideline for Management of Chikanguniya*.

38 National Center for vector Borne Disease Control. (n.d.). *News and Highlights*.

Addressing the needs of vulnerable ageing populations¹

- Increasing elderly population necessitates an expanded availability of elderly care services, encompassing options such as nursing homes, day-care centers, or specialized intensive care tailored to the specific health needs of this demographic.
- It is estimated that more than 30% of elderly individuals live below the poverty line, while an additional 33% are vulnerable to poverty. A majority of these seniors, ~80%, reside in rural areas, and a significant portion, about 73%, lack literacy.



Unmet needs of the urban poor¹

- Healthcare delivery system poses numerous challenges for the urban impoverished population in India. Prevalence of communicable diseases like tuberculosis and NCDs, such as diabetes, is significantly elevated among urban poor communities when contrasted with their rural counterparts.
- ~35% of urban households manage to sustain themselves with less than one dollar per person per day.
- Situation becomes even more complex in disease control efforts due to the substantial population of migrant laborers and daily wage workers in major cities.



Financial burden due to NCDs¹

- Due to shifts in disease patterns leading to an increasing burden of NCDs and injuries in India, healthcare costs have experienced a substantial increase.
- The percentage of out of pocket (OOP) expenses attributable to NCDs increased from 32% in 1995-1996 to 47% in 2004.



Cross Cutting Themes

Climate

- The distribution and prevalence of vector and water-borne diseases are influenced by changing climate conditions.
- Given India's predominantly tropical landscape with heavy rainfall and inadequate water management, rodent and vector-borne diseases worsen seasonally, especially after hydrological disasters.
- Swachh Bharat Mission Promotes sanitation and hygiene practices to reduce the spread of water and vector-borne diseases.

Ministries and Departments

Inter-ministerial convergence among ministries and departments is particularly crucial in the context of disease programmes. Various ministries and departments work together to formulate policies and implement programmes aimed at preventing, controlling, and managing diseases. These collaborative efforts are essential to efficiently address public health challenges.



Ministry of Health
and Family Welfare

Department of Health and Family Welfare

- Department is responsible for implementing a number of disease control programmes in India
- These programmes are designed to prevent, control, and treat a wide range of diseases including communicable, non-communicable diseases, mental health, etc.

Department of Health Research

- The Indian Department of Health Research (DHR) has a variety of illness initiatives in place to enhance the health of the Indian people

National AIDS Control Organisation

- NACO is an Indian government body in charge of organising and implementing the national HIV/AIDS prevention and control programme

Central TB Division

- The Central Tuberculosis Division (CTD) is an Indian Ministry of Health and Family Welfare (MoHFW) division
- It is in charge of carrying out the Revised National Tuberculosis Control Programme (RNTCP) throughout India



Ministry of Social
Justice and
Empowerment

Department of Empowerment of Persons with Disabilities

- The Department of Empowerment of Persons with Disabilities (DEPwD) is an Indian government department in charge of the welfare and development of people with disabilities
- It collaborates with its partners to ensure that people with disabilities have equal access to resources and opportunities



Ministry of Tribal Affairs

Ministry of Tribal Affairs (MoTA) is a ministry of the Government of India that is responsible for the welfare of Scheduled Tribes (STs) in India and implementation of various disease programmes in tribal areas



Ministry of Chemicals
and Fertilizers

Department of Pharmaceuticals

- Department is responsible for the regulation and promotion of the pharmaceutical industry in the country
- It supports research on new diagnostic tests, treatments, and vaccines for treatment of new diseases



Ministry of Science
and Technology

Department of Biotechnology

- Department is responsible for the promotion of biotechnology research and development in the country

Government Interventions

In response to the multifaceted health challenges prevalent within its diverse population, India has formulated a range of comprehensive policies and programmes aimed at tackling various diseases. These initiatives, spanning from infectious diseases to non-communicable ailments, reflect the country's commitment to improving healthcare accessibility and outcomes across its vast landscape.

Prominent national-level programmes are briefly discussed below:

RMNCH+A strategy¹²: The RMNCH+A strategy is constructed around the idea of a continuum of care and is comprehensive in nature, covering all actions targeting reproductive, maternal, newborn, child, and adolescent health within a wide framework. This approach emphasizes a strategic lifecycle perspective, emphasizing the interconnected stages of health and well-being.



Janani Suraksha Yojana

It incentivizes institutional deliveries among pregnant women, especially those from marginalized communities, to reduce maternal and neonatal mortality.



Janani-Shishu Suraksha Karyakram

Aimed at providing free maternal and neonatal healthcare services to pregnant women and newborns, reducing maternal and infant mortality rates.



Pradhan Mantri Surakshit Matritva Abhiyan

To universally offer pregnant women in their 2nd and 3rd trimesters fixed-day, guaranteed, comprehensive, and high-quality antenatal care on the 9th day of each month.



Surakshit Matritva Aashwasan (SUMAN)

To ensure quality healthcare services during pregnancy and childbirth, promoting maternal and child well-being at no cost and zero tolerance for denial.



LaQshya

596 labour rooms (LRS) and 474 Maternity Operation Theatres have attained LaQshya certification for quality service provided for care around birth.



Dakshata

As of 2022, 16,419 health care providers have been Dakshata trained with the focus of high-quality evidence-based practice.

39. NHM. (2018). *Intensified Mission Indradhanush Operational Guidelines*.

Universal Immunization Programme (UIP) is aimed at providing free vaccinations to all eligible children, pregnant women, and other high-risk groups. This programme plays a crucial role in preventing and controlling vaccine-preventable diseases, contributing to the reduction of infant mortality and morbidity rates. Mission Indradhanush was introduced to achieve a full immunization rate of over 90% for newborns by 2020, innovative and strategic methods were employed to ensure the vaccination of all children.³⁹



Intensified Mission Indradhanush Drive

It aims to ensure the consistent delivery of vaccines to target populations, optimize vaccine coverage, and maintain the overall effectiveness of immunization efforts. This process plays a critical role in preventing and controlling vaccine-preventable diseases and safeguarding public health.

Routine Immunization System Strengthening

It focuses on selected districts and urban areas with low immunization coverage and deploys a more concentrated and aggressive approach to ensure that more children and mothers receive the full range of essential vaccines.

Integrated Disease Surveillance Project aims to strengthen disease surveillance and response systems. It operates at both national and state levels, facilitating early detection and containment of disease outbreaks. It utilizes modern technology and epidemiological tools to monitor and control the spread of diseases, contributing significantly to public health management.¹³



Integrated Health Information Platform

To enhance disease surveillance within the nation, a decentralized state-based surveillance system has been designed for diseases prone to epidemics, enabling the prompt detection of early warning signs.



Strategic Health Operations Centre

It is serving as a central hub for real-time data analysis, monitoring disease trends, and coordinating rapid response efforts to address public health emergencies effectively.¹³



Health Emergency Operations Center

It serves as a command and control center where specific emergency management tasks are executed. It assumes responsibility for making strategic and operational choices.⁴⁰

National Tuberculosis Elimination Programme (NTEP) is dedicated to eliminating tuberculosis from the country. As we navigate the journey towards recovery after the pandemic and take inspiration from the "National Strategic Plan (NSP) 2017-2025," the National TB Elimination Programme (NTEP) entered 2022 with renewed vigour, intensifying efforts to achieve its goal of eliminating TB by 2025.¹⁸



TB Diagnostic and Treatment Services

Providing complimentary laboratory services to individuals visiting public healthcare facilities and those referred from the private sector has been the core principle of the programme since its inception. The nation has achieved significant advancements in TB management, encompassing drug-resistant TB (DR-TB) and the widespread adoption of shorter oral regimens throughout the country.



TB Comorbidities and Special Situations

Recognizing the interconnection between TB and undernutrition, the programme has partnered with various ministries to enhance nutrition support for TB-affected individuals, including collaboration with the Ministry of Women and Child Development's "Poshan Mah" initiative, aiming to boost treatment success and nutritional well-being.

40 MOHFW. (n.d.). *Framework for Establishing and Operationalizing State level Health Emergency Operations Centres (HEOC)*.



Supervision and Monitoring

The programme division is enhancing the Ni-kshay portal by integrating advanced analytical tools, enabling the regular provision of crucial indicators' feedback to the states. This will empower them to make essential and timely interventions.



Patient Support System

The programme is enlisted in the Direct Benefit Transfer system, offering a monetary incentive of Rs. 500 per month to each TB patient for the entire duration of their anti-TB treatment. ~₹2,090 crore have been disbursed to ~71 lakh individuals impacted by TB through the Ni-kshay Poshan Yojana (NPY) between April 2018 and December 2022.



Pradhan Mantri TB Mukht Bharat Abhiyaan

To offer TB-affected individuals and their families supplementary nutritional, diagnostic, and vocational assistance through community-based delivery, this initiative has received an overwhelming response since its inception. As of 1 January, 2023, over 58,000 Ni-kshay Mitras (donors) have stepped forward, pledging their support to over nine lakh consenting TB-affected individuals.



Advocacy, Communication and Social Mobilization Including Community Engagement

It has been fortifying the programme's base by strengthening efforts in all aspects of TB care. To empower TB survivors with fundamental knowledge about TB and the services provided by the NTEP, a certificate course called "Self-learning course for TB Champions" has been created and made accessible on various e-platforms.

National AIDS Control Organization (NACO) is responsible for formulating and implementing policies and programmes to combat HIV/AIDS. NACO plays a central role in coordinating efforts to prevent the spread of HIV, provide treatment and care to affected individuals, and reduce the stigma associated with the disease.¹²



Campaign on Stigma and Discrimination

A nationwide initiative (#AbNahiChalega) was inaugurated on World AIDS Day 2022 to combat and eradicate HIV-related stigma and discrimination within society.



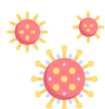
Adolescence Education Programme

It is a vital intervention aimed at enhancing the life skills of young individuals, enabling adolescents to effectively manage peer pressure, and fostering positive behaviors by promoting awareness of safe sexual health to prevent HIV transmission.



Red Ribbon Club

It is a promotional and preventative initiative designed to tap into the potential of youth within educational institutions, with a particular focus on integrating NACP services among young people and promoting voluntary blood donation.



TB-HIV linkage activities in Targeted Intervention Programme

During their regular outreach services, either the peer educator (PE) or other team members screen high-risk groups (HRGs) and bridge populations for TB symptoms and subsequently refer them to the nearest TB center for confirmation and treatment.



Opioid Substitution Therapy (OST) Programme for IDUs

It entails the treatment of opioid-dependent individuals who inject drug using a long-acting opioid agonist medication administered through the sublingual route over an extended period. It effectively reduces cravings and withdrawal symptoms, helping IDUs discontinue the practice of injecting drugs.



Sampoorna Suraksha Strategy

The strategy aims to prevent HIV transmission among individuals who don't fit the traditional definition of Key Population, as they have a negative HIV status but face elevated risk due to their risky behavior or that of their spouse/partner(s). This involves offering them a periodic, tailored, and comprehensive package of supportive services designed to help them maintain their negative status and overall well-being.

National Vector Borne Disease Control Programme: It is dedicated to combating diseases spread by vectors such as mosquitoes. focuses on the prevention and control of vector-borne diseases like malaria, dengue, chikungunya, and lymphatic filariasis through measures such as vector control, surveillance, and health education.¹²



Monthly Malaria Information System (MMIS)

It serves as a data-driven platform that compiles and analyses monthly malaria-related information, aiding health authorities in monitoring disease trends, planning interventions, and ensuring timely and targeted malaria control measures, ultimately contributing to the reduction of malaria cases and deaths.



Established Sentinel Surveillance Hospitals

It selected healthcare facilities equipped to monitor and report specific to dengue in endemic states, providing valuable data for disease surveillance and early detection of outbreaks. All of these institutions are connected to 17 Apex Referral Laboratories that offer comprehensive diagnostic capabilities, providing essential backup support.

DRAFT

Disease Programme Milestones



Prevalence rate for leprosy has decreased to 0.57 per 10,000 population (data for January 2022 to September 2022.)¹²



In 2022, a significant milestone was achieved in India's TB surveillance efforts, with a remarkable record high of 24.2 lakh cases reported, representing a growth of more than 13% compared to 2021.¹⁷



Integrated Information Management System is NACP's central data reporting platform, a patient-centered, integrated IT system crucial for digital transformation and achieving the global HIV target of 95-95-95.⁴¹



After introduction of National Tobacco Control Programme number of tobacco users has reduced by ~81 lakh between 2009-2010 to 2016-2017.¹²



India has achieved a reduction of 92.03% in malaria morbidity and 90.34% in malaria mortality in 2021 as compared to 2020.⁴³



National Leprosy Eradication Programme reduced the prevalence rate of leprosy from 0.69 per 10,000 population in 2014-2015 to 0.45 in 2021-2022.⁴³

⁴¹ NACO. (2022). *NEWS April-June 2022*.

⁴² MOHFW. (n.d.). *National Tobacco Control Programme*. Retrieved 8 September, 2023.

⁴³ World Health Organization. (2022). *Supporting leprosy elimination in India*.

Way Forward

Disease Surveillance



- Integrated Health Information Platform (IHIP) has demonstrated potential to detect epidemics, issue early warning signals in few states.³⁵
- Similar research and surveillance programmes is being scaled to incorporate more states and regions.

Digital Intervention



- People residing in remote and tribal areas access to city-level medical facilities owing to the expanding National Telemedicine Service.
- Rapid development of point of care tests and hand-held devices is helping to reach populations that are otherwise unreachable by the health system and can facilitate timely diagnoses.³⁵

Community Participation



- Engagement of community representatives like TB champion for National TB programme ensures success of the disease programme.¹⁷
- Such, community Engagement will continue to be key to improve health service coverage and deliver accessible and people-centred integrated care.

1. Improving Access to Malaria Control Services - Odisha: A mission to eradicate malaria from the state of Odisha by making ASHA workers more active in the

1. Background:

Malaria control in the Indian state of Odisha is beset by significant challenges. Numerous remote and tribal regions within Odisha are situated in rugged terrains, rendering it arduous for healthcare workers and facilities to access these populations. Additionally, Odisha encounters seasonal surges in malaria cases, which can strain healthcare facilities and resources. The intricate biology of malaria vectors in Odisha further complicates efforts to curb malaria transmission. Alarming, Odisha exhibits a remarkably high prevalence of *Plasmodium falciparum*, the most severe and life-threatening malaria parasite, accounting for over 90% of malaria cases in the region.

2. Steps taken by the Government:

- **DAMaN – Access the Inaccessible:** The State of Odisha launched a pioneering initiative in 2017, the first of its kind in India. This initiative utilizes a camp-based approach with two annual rounds of implementation. It focuses on the most vulnerable populations and involves the participation of ASHA workers, Anganwadi Workers (AWW), along with other community volunteers and members of the Panchayati Raj Institutions (PRI). The initiative includes comprehensive malaria screening, targeting both symptomatic and asymptomatic cases. All individuals testing positive for malaria receive treatment. Additionally, it involves the estimation of haemoglobin levels in pregnant and lactating mothers, as well as growth assessment for children under the age of five.
- **ASHA and Community Volunteers:** The mandatory malaria screening is carried out using a bi-valent Rapid Diagnostic (RD) test. Positive cases are treated in accordance with national guidelines. Follow-up measures are implemented to ensure treatment adherence and monitor any potential complications. Additionally, counselling sessions are conducted to promote better acceptance and utilization of vector control interventions. Information about these activities is documented on the village health bulletin known as the "Swasthy Kanth." Furthermore, meetings are organized within the village community to discuss the distribution and proper usage of Long-Lasting Insecticidal Nets (LLINs).

3. Impact generated:

- Despite an increase in case detection, there was a remarkable 187% reduction in deaths in 2016 as compared to the numbers recorded in 2007.
- Between 2010 and 2012, ~40 lakh LLINs were successfully distributed through the ASHAs-GKS mechanism, leading to a significant reduction in malaria incidence and deaths. In 2013, these reductions amounted to 73.54% for incidence and an impressive 268.7% for deaths when compared to the numbers recorded in 2010. Early diagnosis and complete treatment, mass screening in hard to reach areas and Tribal Residential Schools averted outbreaks.
- Increase in Passive Surveillance-increase from 12.24% in 2007 to 16.37% in 2016.
- In 2017, >11.34 million LLINs are being distributed under The Global Fund to Fight AIDS, Tuberculosis and Malaria support - already seven million have been distributed through ASHA-GKS mechanism within two months and most difficult pockets have been protected before cut off by rains.

LLNs: Long Lasting Insecticidal Nets

⁴⁴ NHSRC: *Malaria Control Services, Odisha*.

⁴⁵ NHSRC: *Orientation of ASHAs for NCDs, J&K*.



Fig 2- Malaria Control Services

2. Screening and Orientation of ASHAs for NCDs – Jammu and Kashmir: FOR NON-COMMUNICABLE DISEASES AT PULWAMA DISTRICT⁴⁵

1. **Background:** The National Health Mission (NHM) in Jammu and Kashmir made the decision to evaluate the health status of ASHAs and initiated a screening programme to assess them for non-communicable diseases such as diabetes, hypertension, thyroid issues, dyslipidemias, PCOD, and obesity. To carry out these health check-ups for ASHAs, who play a crucial role in the operational components of the '4 A Platform,' distinguished faculty members from AIIMS and SKIIMS were tasked with conducting these assessments at the ASHAs' doorstep. This initiative was driven by the Health Department and NHM's desire to foster a sense of care and well-being among these healthcare cadres. The comprehensive day-long testing and check-up sessions provided ASHAs with practical experience and a deeper understanding of non-communicable diseases. It served as a vital preparatory exercise for the subsequent rollout of population-based screening for non-communicable diseases, as well as their management by these cadres. The plan was to expand this initiative to all 22 districts of the state, enabling the study of the distribution of non-communicable diseases across different geographical areas within the state.
2. **Steps taken by the Government:** NHM, Jammu and Kashmir worked in coordination with Faculty of Endocrinology, AIIMS New Delhi and SKIIMS, Srinagar to take up a pilot district. The following factors were considered: -
 - Selection of the District.
 - Adequate and timely IEC.
 - No. of the ASHAs to be screened to determine the logistics.
 - Senior Faculty of Endocrinology, from AIIMS, New Delhi, along with Professor of Radiology and concerned laboratory personnel in adequate number from SKIIMS Kashmir screened the ASHAs of Pulwama District.
 - The budget for investigations was shared by SKIIMS and the budget for refreshment and local logistics was borne by District Health Administration, Pulwama.
 - Orientation of ASHAs on NCDs: On the same day they were also oriented on Non-communicable diseases in a comprehensive session in the local language by the AIIMS Faculty and District Health Administration, Pulwama.
 - Grievance Redressal: - To add value from perspective of ASHAs, a comprehensive Grievance Redressal was held by Mission Director, NHM Jammu and Kashmir and District Magistrate and District Health Authorities, Pulwama.

3. Impact generated:

- Improved morale of ASHAs, mutual trust and bonding between ASHAs and the department.
- ASHAs acted as ambassadors for programme of prevention of NCDs in their respective community.
- Orientation on NCDs followed by the screening increased the receptiveness of ASHAs about importance of timely NCD screening. This will have positive impact on her functionality to mobilize community.
- Screening of other allied healthcare workers and ASHAs across the state can be conducted.



Fig 3- Orientation for ASHA

3. Corneal Blindness Backlog Free Punjab: To eradicate corneal blindness⁴⁶

1. Background:

A pivotal shift occurred as corneal blindness was acknowledged as a significant issue warranting focused attention, marking the Government of Punjab as the pioneer among state governments to initiate a mission to eliminate the backlog of corneal blindness cases. In practical terms, it was recognized that waiting lists for treatment are continually changing, but the primary objective was to address the backlog of corneal blind patients who could benefit from keratoplasty. The estimation was that approximately 1,400 patients required treatment, and to account for variations, it was assumed that an average utilization rate of 50% would be achieved. Consequently, a target of arranging 3,000 corneal donations was set.

2. Steps taken by the Government:

In September 2015, the Punjab Health Department launched the "Corneal Blindness Backlog Free Punjab (CBBF)" initiative, aiming to eliminate corneal blindness through a comprehensive strategy known as DREAM (Digitization, Re-energization, Education and Empowerment, Advocacy, and Multitasking). This approach engaged all stakeholders, fostering collaboration between government agencies, professional eye banks, private institutions, and non-governmental organizations. Together, they worked to assess the approximate number of corneal blind patients in need of transplant procedures.

The compilation of the list was achieved by consolidating data from multiple sources, including waiting lists from state eye banks, patients screened at outpatient departments (OPDs) and eye camps, and the training of healthcare workers, including ASHAs, to identify individuals with visual disabilities and refer them to healthcare facilities. To enhance coordination and minimize corneal tissue wastage, a "WhatsApp Group" was established, connecting corneal transplant surgeons, eye banks, and social workers. This group facilitated direct communication between corneal surgeons and the organizations and institutions responsible for retrieving corneas. Regular review meetings were conducted, and any necessary corrective actions were taken promptly to address issues as they arose.

Awareness Van is launched which visits the major towns of all the districts of the State. This van carries eye donation cards. These cards are filled and a registry of eye donors in the State is being maintained.

3. Impact generated:

Eye Care Services	Target year 2017-18	Achievement for the year 2017-18
Eye Donations	700	1,274

- There is no bilateral blind patient awaiting keratoplasty for a long time in Punjab at present.
- At present, there are 178 patients in the waiting list of State. All these are fresh cases.
- Patients have to wait for surgery due to shortage of good quality cornea.
- Approx 250 corneal blind patients from Punjab are waiting for surgery in PGIMER Chandigarh.
- A community-based identification of cases with referral, mobilization of resources and surgery was able to achieve the desired results.
- The state is starting new keratoplasty centre at Civil Hospital, Patiala to further enhance the number of keratoplasties.



Fig 4- Awareness Van

4. Integrated Approach to Elimination of Grade II Disability due to Leprosy- Dadra and Nagar⁴⁷ Haveli

1. Background:

The key factor behind stigma associated with Leprosy is the deformities/ grade II (visible) disabilities leading to permanent disfigurement. Probable reasons of high endemicity:

- Environmental Conditions – Hot and Humid for most part of the year.
- Housing – Poorly ventilated, kachcha floorings and dark interiors (no windows).
- Tribal population – Leprosy in DNH mostly in Tribals (98%).
- Nutritional status/ Immunity/ Genetic predilection of Tribal Population in that area.
- Adjoining High endemic blocks – Gujarat and Maharashtra.

2. Steps taken by the Government:

- Detect Leprosy cases (including hidden cases) early before development of any disability.
- Halt active transmission by early diagnosis and prompt treatment.
- Decrease Prevalence Rate and Annual New Case Detection Rate in the long term (three – five years) because newer case leads to more likelihood of G2D.
- Three rounds of Active Case Finding Campaigns and one round of Focused Leprosy Campaign for early detection.
- Active Case Detection Survey twice by General Health Staff.
- Active search in evening hours (6 - 8 PM) to screen working population.
- Sensitized and Trained all the Medical Officers, Supervisors, ANM, MPWs and ASHAs to screen households for sensory loss in addition to hypo-pigmented patch and visible deformity.
- Increased number of households for contacts screening from 25 to 50 or more depending on adjoining population for every new case.
- Started contact tracing of PB cases also.
- Started follow up of Released From Treatment cases up to five years.

⁴⁶ NHRSC. (n.d.). *Corneal Blindness Backlog Free Punjab*.

⁴⁷ NHRSC. (n.d.). *Grade II Disability, Dadra and Nagar Haveli*.

3. Impact generated:

- In the 2017-18, only one case with Grade II disability was verified, and this resulted in the most substantial reduction in the percentage of Grade 2 disability.
- G2D at the time of diagnosis was reduced from 2.08% in 2016-17 to a remarkable 0.36% in 2017-2018 and in , 2018-19: Reached the target of “Zero Grade II Disability” due to Leprosy till now.



Fig 5- Leprosy Check-ups

Recommended Readings

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